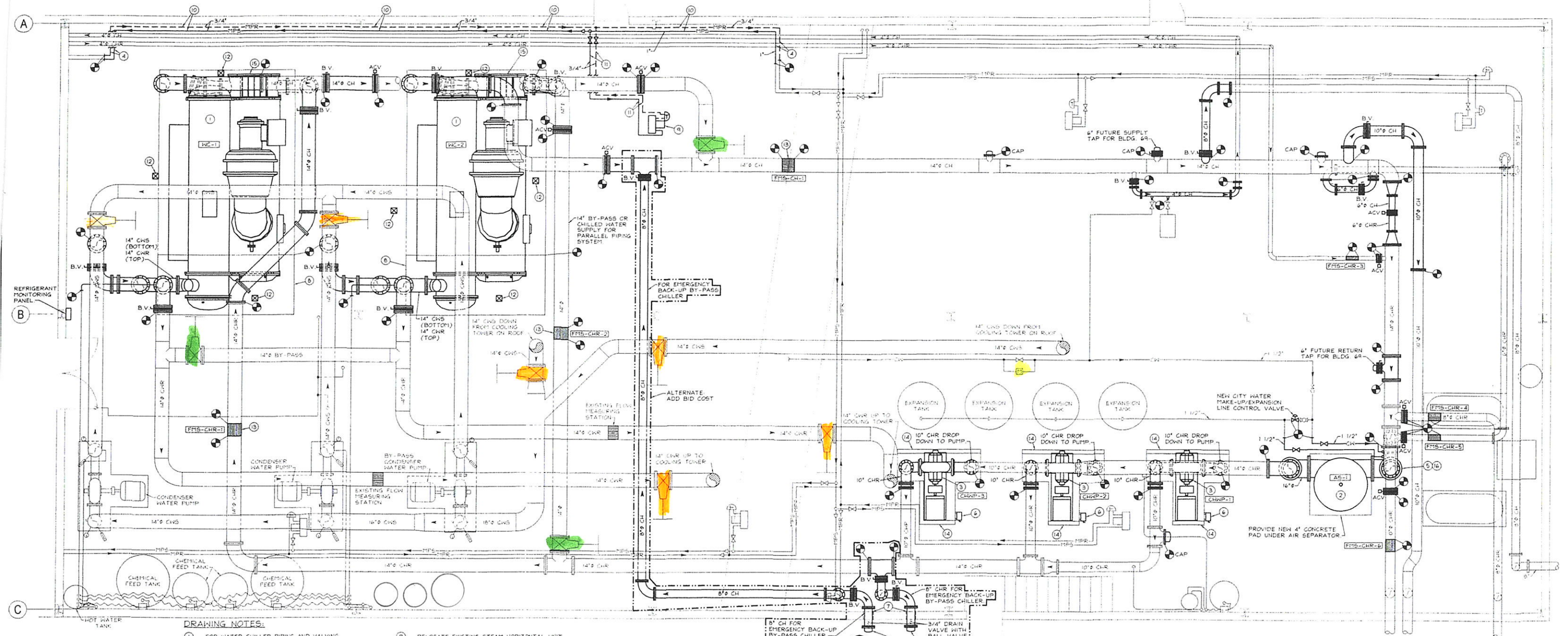


3 water meters w Remote readers



HVAC NEW WORK PLAN
SCALE: 1/4"=1'-0"

- ### DRAINING NOTES:
- | | | | |
|---|---|---|--|
| ① | FOR WATER CHILLER PIPING AND VALVING INSTALLATION, SEE DETAIL  | ⑨ | RELOCATE EXISTING STEAM HORIZONTAL UNIT HEATER, PROVIDE NEW VALVES, TRAP, AND SUPPORTS FOR UNIT. EXTEND AND RECONNECT EXISTING POWER AND CONTROL WIRING TO NEW LOCATION. |
| ② | FOR AIR SEPARATOR INSTALLATION, SEE DETAIL  | ⑩ | RELOCATED PIPES AND MPR PIPING PROVIDE NEW HANGERS AND SUPPORTS. |
| ③ | FOR HORIZONTAL SPLIT CASE WATER PUMP INSTALLATION, SEE DETAIL  | ⑪ | INSTALL NEW MPS AND MPR PIPING TO RELOCATED UNIT HEATER. INSULATE PIPING WITH KRAFT FACED, 1 1/2" THICK, FIBERGLASS, PIPE INSULATION. |
| ④ | PROVIDE NEW SECTION OF PIPING TO ALLOW EXISTING MPS & MPR PIPING TO BE SHIFTED TO BACK WALL. | ⑫ | REFRIGERANT SENSOR, CONNECT TO NEW REFRIGERANT MONITORING SYSTEM (TYPICAL) |
| ⑤ | PROVIDE PIPE STANCHION SUPPORT UNDER ELBOWS FOR VERTICAL PIPING SUPPORT. PROVIDE CONCRETE PAD AND FLOOR PLATE FOR ANCHOR AND SUPPORTING PIPING. (TYPICAL) | ⑬ | INSTALL NEW FLOW MEASURING 5" ACTION IN EXISTING CHILLED WATER PIPING. (TYPICAL) |
| ⑥ | VARIABLE FREQUENCY DRIVE FOR PUMPS. PROVIDE STRUCTURAL UNISTRUT FOR MOUNTING DRIVES NEXT TO PUMPS. (TYPICAL) | ⑭ | EXTEND CONCRETE PAD AS REQUIRED FOR INSTALLATION OF NEW PUMP. HEIGHT OF NEW PAD SHALL MATCH THAT OF EXISTING PUMP PAD. PROVIDE DOWELS FOR FASTENING TO EXISTING PAD AND FLOOR. |
| ⑦ | CORE DRILL WALL AND PROVIDE SLEEVES, SEAL PENETRATION TO MAINTAIN A WEATHER TIGHT SEAL. | ⑮ | FOR CHILLER EVAPORATOR END VIEW, SEE DETAIL  |
| ⑧ | PROVIDE 4" HIGH CONCRETE PAD TO MATCH EXISTING CHILLER PAD. PROVIDE DOWELS FOR FASTENING TO EXISTING PAD AND FLOOR. | ⑯ | TRANSITION FROM 16" DIA. TO 14" DIA. IN RISE |



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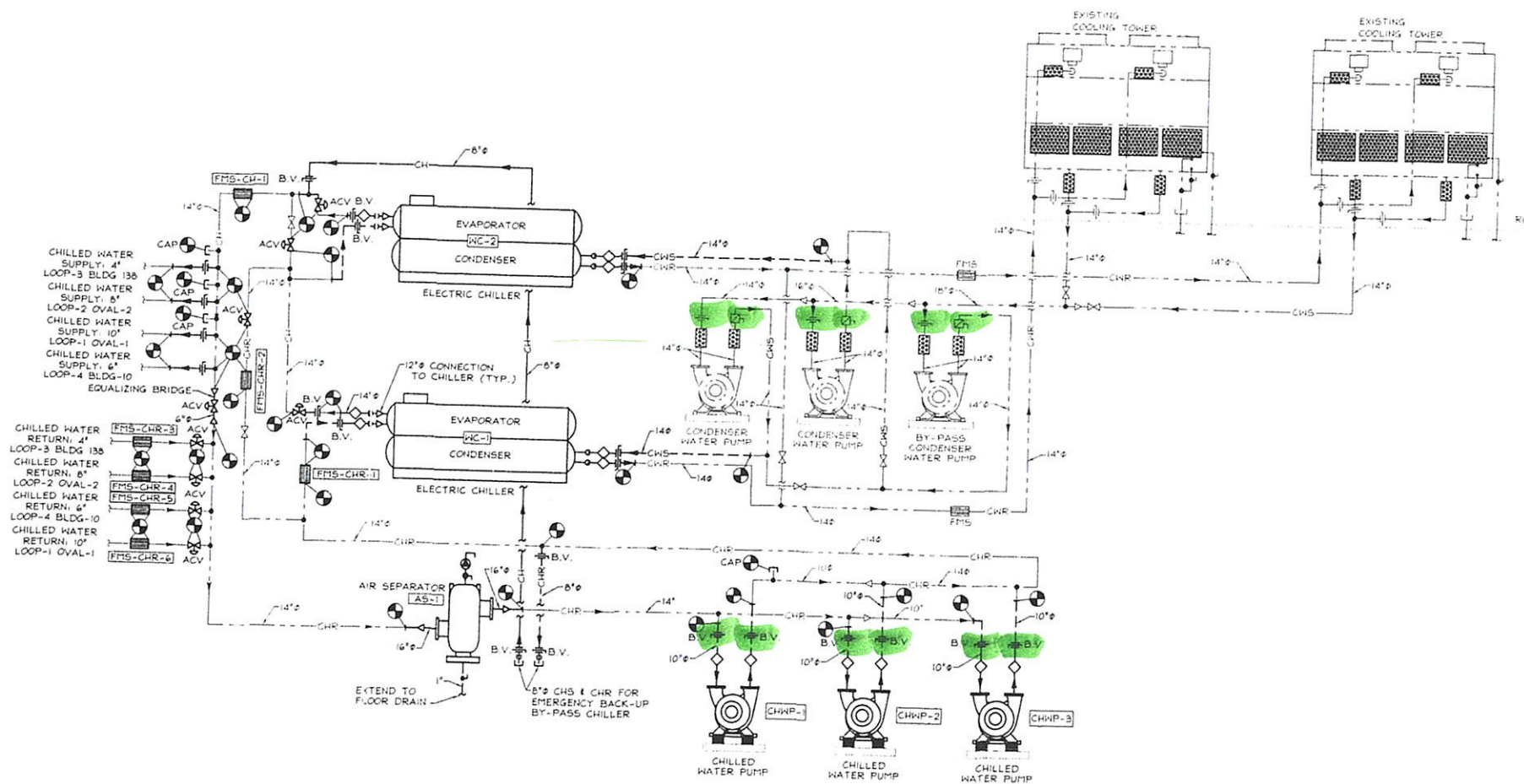
 **FIRSCHING,
MARSTILLER,
RUSBARSKY AND
WOLF**

ENGINEERING, INC. 

2030 ARDMORE BOULEVARD, SUITE 250
PITTSBURGH, PA 15221
(412) 271-5090 PHONE (412) 271-5193 FAX
E NO. 07062-1

DRAWING TITLE HVAC NEW WORK PLAN		PROJECT TITLE CHILLER REPLACEMENT		DATE 18 JAN. 2008	DEPARTMENT OF VETERANS AFFAIRS
				SCALE AS NOTED	
PROJECT NO. XXX-XX-XXX	CONTRACT NO. V244C-00236	DESIGNED BY: DDR	DRAWN BY: DSD	CHECKED BY: CEM	
BUILDING NO. 1	AUTOCAD FILE NAME X-1.DWG	LOCATION VAMC COATESVILLE			DRAWING NO. H-300

DEPARTMENT OF
VETERANS AFFAIRS



CHILLED & CONDENSER WATER FLOW DIAGRAM
NOT TO SCALE



DEPARTMENT OF
VETERANS AFFAIRS

DATE
18 JAN. 2008

SCALE
AS NOTED

DRAWING NO.
H-400

PROJECT TITLE		
CHILLER REPLACEMENT		
DESIGNED BY: DDR	DRAWN BY: DSD	CHECKED BY: CEM

DRAWING TITLE	
CHILLED & CONDENSER WATER FLOW DIAGRAM	
PROJECT NO. XXX-XX-XXX	CONTRACT NO. V244C-00236
BUILDING NO. 1	AUTOCAD FILE NAME Y-1.DWG
LOCATION HANS COATESVILLE	

FIRSCHING, MARSTILLER, RUSSEBARSKY AND WOLF
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Coatesville
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REVISIONS	DATE

WATER CHILLERS

TAG NUMBER	NOMINAL CAPACITY (TONS)	EVAPORATOR				CONDENSER				NO. OF COMP.	ELECTRICAL				DESIGN BASIS
		GPM	MAX DP (FT.)	ENT (°F)	LWT (°F)	GPM	MAX DP (FT.)	ENT (°F)	LWT (°F)		VOLTS/PHASE	KW/TON	MOTOR RLA	MICA	
WC-1	1475	3040	29.85	58.74	49.70	3000	18.08	85	95.02	1	460/3	0.511	89.5	114	587.90
WC-2	1475	3040	29.7	49.70	42.00	3000	18.08	85	93.63	1	460/3	0.561	84.2	107	549.50

NOTE: EACH CHILLER REQUIRES 1700 LBS. OF HCFC-123 REFRIGERANT.

PUMPS

TAG NUMBER	TYPE	LOCATION	GPM	HEAD (FT.)	NPSH (FT.)	MIN. EFF.	PUMP TEMP (°F)	MOTOR				IMPELLER SIZE	MANUFACTURER	MODEL
								RPM	HP	VOLTS/PHASE				
CHWP-1	DOUBLE SUCTION	MECH. ROOM	1800	200	-	87	45	1750	125	460/3	13.625	PACO	6015-1/2 (6x8.15)	
CHWP-2	DOUBLE SUCTION	MECH. ROOM	1800	200	-	87	45	1750	125	460/3	13.625	PACO	6015-1/2 (6x8.15)	
CHWP-3	DOUBLE SUCTION	MECH. ROOM	1800	200	-	87	45	1750	125	460/3	13.625	PACO	6015-1/2 (6x8.15)	

*STAND-BY PUMP

HYDRONIC FLOW MEASUREMENT STATION

TAG NUMBER	TYPE	SYSTEM SERVED	CHILLERS	WATER TEMP. (°F)	LINE SIZE (INCHES)	GPM/PPM		MANUFACTURER
						MINIMUM	(NOMINAL) MAXIMUM	
FMS-CHR-1	IMPELLER	CHILLED WATER	WC-1	55	14	0	5000	DATA INDUSTRIAL
FMS-CHR-2	IMPELLER	CHILLED WATER	WC-2	55	14	0	5000	DATA INDUSTRIAL
FMS-CHR-3	IMPELLER	CHILLED WATER	WC-1 & 2	40	14	0	5000	DATA INDUSTRIAL
FMS-CHR-4	IMPELLER	CHILLED WATER	WC-1 & 2	55	4	0	65	DATA INDUSTRIAL
FMS-CHR-5	IMPELLER	CHILLED WATER	WC-1 & 2	55	8	0	650	DATA INDUSTRIAL
FMS-CHR-6	IMPELLER	CHILLED WATER	WC-1 & 2	55	6	0	170	DATA INDUSTRIAL
FMS-CHR-6	IMPELLER	CHILLED WATER	WC-1 & 2	55	10	0	900	DATA INDUSTRIAL

AIR SEPARATORS

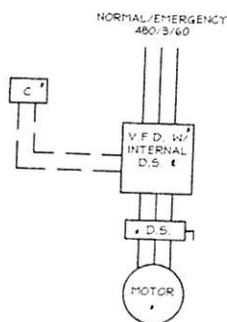
TAG NUMBER	GPM	CON. SIZE (INCHES)	STRAINER FREE AREA (SQUARE INCHES)	SYSTEM SERVED	MANUFACTURER	MODEL
AS-1	6,400	16	1,517	CHILLED WATER	AMTROL	18-AS

ELECTRICAL NOTES:

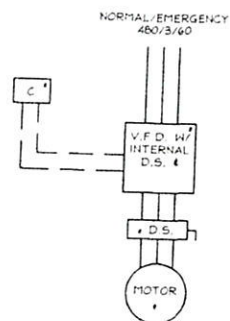
- ALL STARTERS, DISCONNECTS, CONTROL DEVICES MARKED WITH AN (*) ASTERISK SHOWN ON THESE WIRING DIAGRAMS SHALL BE FURNISHED BY THE HEATING CONTRACTOR (H.C.). SEE SPECIFICATIONS FOR DEFINITIONS OF THE WORDS FURNISHED, INSTALLED, PROVIDED FOR THIS PROJECT.
- ALL POWER WIRING SHALL BE PROVIDED BY THE ELECTRICAL CONTRACTOR.
- CONTROL WIRING IS SHOWN BY DASHED LINES TO THE CONTROL DEVICES. ALL CONTROL WIRING SHALL BE FURNISHED BY THE AUTOMATIC TEMPERATURE CONTROL (ATC) SUBCONTRACTOR.
- ALL MOTOR HORSEPOWERS ARE SHOWN IN THE EQUIPMENT SCHEDULE. CONTRACTOR SHALL VERIFY THAT HORSEPOWERS OF EQUIPMENT SUPPLIED MATCH THAT OF EQUIPMENT SCHEDULED WITHOUT ANY DISCREPANCIES. THE CONTRACTOR SHALL CONTACT THE ENGINEER FOR SIZE VERIFICATION DURING BIDS.
- ALL WIRING (INCLUDING LOW VOLTAGE CONTROL WIRING) SHALL BE RUN IN EMT CONDUIT EXCEPT FOR FINAL CONNECTIONS TO ROTATING OR VIBRATING EQUIPMENT WHICH SHALL BE MADE WITH FLEXIBLE GREENFIELD CONDUIT-NOT TO EXCEED 18' IN LENGTH.
- ALL WIRING SHALL CONFORM TO THE SPECIFICATIONS AND TO THE NATIONAL ELECTRIC CODE.

ELECTRICAL ABBREVIATIONS

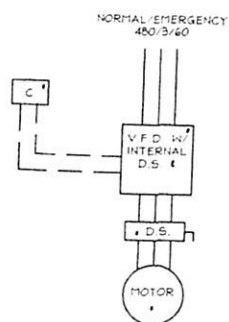
D.S. DISCONNECT SWITCH
C CONTROL
V.F.D. VARIABLE FREQUENCY DRIVE



CHILLED WATER PUMP
CHWP-1
125 HP
(FREE STANDING)

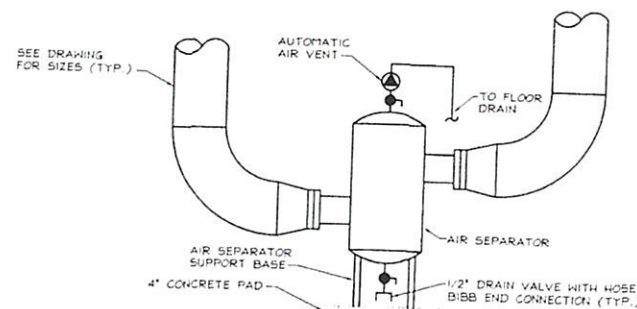


CHILLED WATER PUMP
CHWP-2
125 HP
(FREE STANDING)

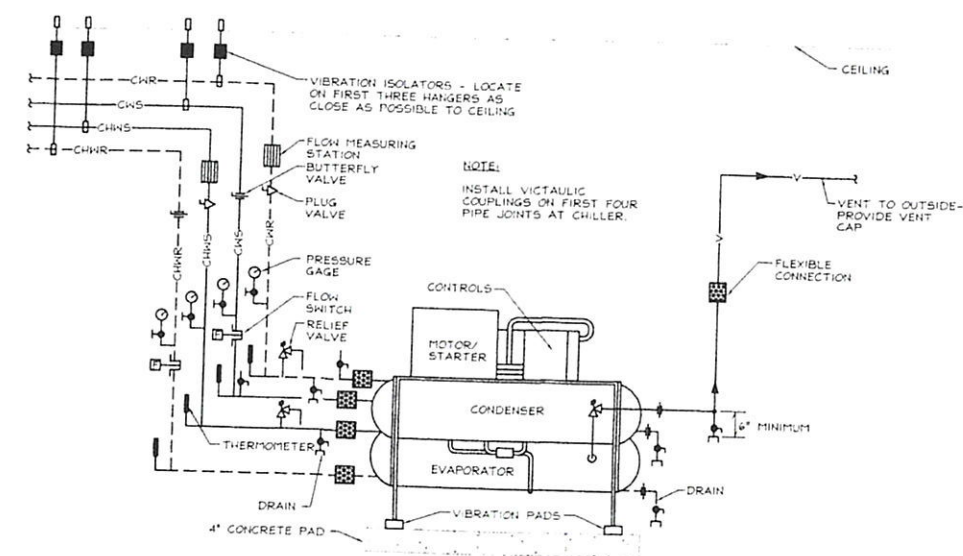


CHILLED WATER PUMP
CHWP-3
125 HP
(FREE STANDING)

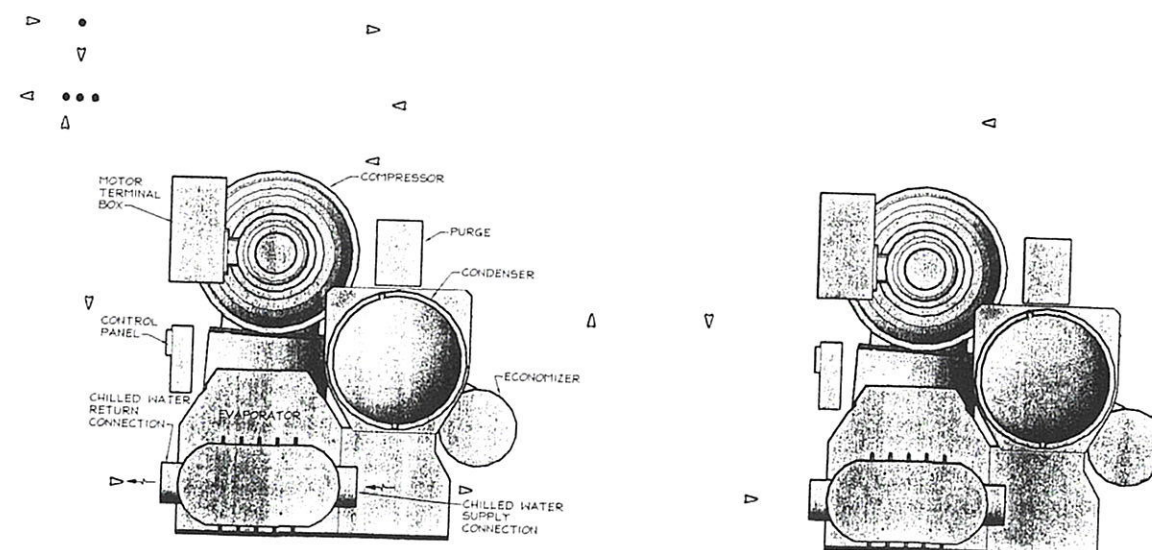
1 HORIZONTAL SPLIT-CASE PUMP
H-500
DETAIL
NOT TO SCALE



3 AIR SEPARATOR
H-500
DETAIL
NOT TO SCALE



2 ELECTRIC CHILLER PIPING DIAGRAM
H-500
DETAIL
NOT TO SCALE



4 CHILLER END DETAIL
H-500
DETAIL
NOT TO SCALE



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FILE NO. 07062-1

DRAWING TITLE HVAC SCHEDULES, DETAILS AND WIRING DIAGRAMS		PROJECT TITLE CHILLER REPLACEMENT		DATE 18 JAN 2008
PROJECT NO. XXX-XX-XXX	CONTRACT NO. V244C-00236	DESIGNED BY: DDR	DRAWN BY: DSD	CHECKED BY: CEM
BUILDING NO.	AUTOCAD FILE NAME	LOCATION	DRAWING NO. H-500	

DEPARTMENT OF
GRANS AFFAIRS